ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. **594 out of 2257**
2. **Frequent pattern mining discovers recurring itemset in a dataset. In order to have a good result from this mining method, each item in a dataset should be comparable. Therefore, isolating specific categories is a crucial first step here.**

AI used for this answer: Yes / **No**

If yes, how:

External references (links):

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Because dropping that much of data would also drop other important data and the remaining 60% might be bias. So, to make text data can still be utilized, we can replace missing values with estimated values. Some cases can't use this method such as sensitive data.

AI used for this answer: Yes / **No**

If yes, how:

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, the largest gap between the value in the sample and in the original was 1.92 percentage.
2. Simple random sampling may be bias and completely miss minority classes.

AI used for this answer: Yes / No

If yes, how: for generating code to test proportion between sample and original

External references (links): https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.sample.html

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The stop_words='english' shape=(2257, 35482)
   stop_words=None shape=(2257, 35788)
   So the total number of features increase
2. Adding stopwords back decreases the overall sparsity because these common words appear in many documents which cause more entries to become non-zero.

AI used for this answer: Yes / **No**

If yes, how:

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

1. The vocabulary drops from 35,482 to 5,089 to 1,491 terms.
2. Rare words that appear in very few documents which may be jargons or slangs, because they are 0 almost everywhere.
3. Because they carry little information that a model can use. Removing them is similar to removing noise which enhance model speed.

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes / **No**

If yes, how:

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

3. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
4. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
5. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Most top words are domain-specific for graphics (vga, video, card, image, 3d), with a few generic newsgroup words like thanks, hi and nntp.
2. Pearson measures how well the raw values follow a straight line. Spearman does the same calculation on ranks, so it measures any consistent increase.
3. When a word's count rises with the topic but not linearly, or when a few extreme counts distort the raw values.

AI used for this answer: Yes / No

If yes, how: To create a code displaying top 10

External references (links):
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup: The pattern count rises from 710 to 9,547, and the runtime rises from about 0.1001 s to about 2.8089s.

Answer (2) — Top-K k vs MaxFPGrowth minSup: k fixes how many patterns you get, while minSup fixes the minimum support each pattern must reach.

Answer (3) — Maximal vs standard frequent patterns: Because maximal pattern has no frequent superset.

AI used for this answer: Yes / **No**

If yes, how:

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. The raw matrix has 35,482 axes (one per word), but a screen shows only 2 (or 3). The data can't be drawn directly, so each document must first be reduced to 2–3 numbers.
2. Dimensionality reduction, including PCA, t-SNE and UMAP, which visualization_helpers.py uses

AI used for this answer: Yes / No

If yes, how: to recheck the concept

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. t-SNE/UMAP show separated clusters. PCA shows heavy overlap in the center.
2. PCA is linear and keeps only the 2 directions of largest variance. Text categories can't be split along straight lines. t-SNE/UMAP keep neighbourhoods, so similar documents stay together.
3. t-SNE does not preserve global distances. The gaps and sizes between clusters can change with settings and may mean nothing.

AI used for this answer: Yes / No

If yes, how: to recheck the concept

External references (links): https://distill.pub/2016/misread-tsne/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?

2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 binary columns, one per category.
2. It assumes the categories have an order and equal spacing. For example, category 3 is "bigger" than 1, and the distance between 0 and 3 is three times the distance between 0 and 1. But the categories are just names with no order.
3. One-hot gives each category its own 0/1 column. No category is larger than another, and every pair is equally far apart, so the false order disappears.

AI used for this answer: Yes / No

If yes, how: to recheck the concept

External references (links): https://scikit-learn.org/stable/modules/preprocessing.html#encoding-categorical-features

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